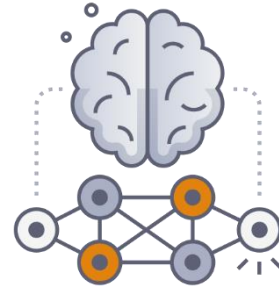


CSC3331

Introduction to Big Data Environment and Applications

Fall2025

Project1: Knowledge Graph



- This is your first CSC3331 project that counts 10% toward your final grade
- This project is assessing ILO5: **Be able to write and run an application using a big data environment**
- Main skills acquired after finishing the project:
 - o Being familiar with Optical Character Recognition (OCR) using ChatGPT
 - o Work with a document database: using MongoDB
 - o Work with a graph database: using Neo4j
 - o Use Python as a programming language
- Except for 1 student, this project is a team-based project where 1 project team member must submit a video (not a link to a video) explaining the project steps
 - o Keep an eye on bold statements
 - o Project is due Monday October 27th before midnight
 - Video must be uploaded via Canvas
 - Your video should not exceed 30 minutes (Otherwise, I will apply a penalty of -10%)
 - No project is accepted if you miss the deadline
 - Projects submitted not through Canvas will not be graded
 - When you present your work, make sure that the font is visible and can be read clearly
 - As you present a code/query, don't read the content but explain the rational behind it
- Any form of plagiarism will imply an immediate WF as a course grade and a report will be submitted to the disciplinary committee in the next 24 hours after grading the work.

Let's get started:

Part 1: Pictures Time!

- You will use the university catalog found in this link:
<https://aui.ma/hubfs/AUI%20Academic%20Catalog%202024-2025.pdf?hsLang=en>
- BSCSC degree consists of **six areas** that you can find on page 193
- Each team will take care of **generating/verifying clear** screenshots for the courses' descriptions in each area.

- o Use PNG as picture extension
- o An example of a clear picture:

CSC 3331 Introduction to Big Data Environment and Application (3 SCH)

Prerequisite: CSC 3326

3 lecture hours

The course introduces Big Data management and techniques that can be applied to massive datasets in distributed environments. The course covers the Map-Reduce parallel computing paradigm and Hadoop distributed file system. The course reviews data storage and preparation for applications, including some machine learning algorithms used for mining knowledge in datastores, including NoSQL.

- o Courses descriptions can be found at the end of the catalog (starting page 221)
- o These screenshots should be saved in the appropriate folders that you can find in the following link: [CSC3331_Project1_Files](#)
 - Deadline to finish this Task is: **Tuesday October 21st before midnight**
- o **Some Teams will have to detect anomalies; A chosen member will have to contact me by email Wednesday October 22nd before 15:00.**
 - A penalty of 10% will be applied to teams' work with anomalies
- o Each generated picture should be **named the same as the course code**. Please put the course code discipline in capital letters (CSC3331 not csc3331).
- o Screenshots work is sharded as follows (**Please read comments in the table below**):

(Oussama Oukkassou, Ilyas Bartal)
Comment :
 When you refer to General Education Folder, create a subfolder for each discipline:
 English, Arabic,

First Year Experience	FYE 1101 + FYE 1102	2
Foundations for Academic Success	FAS 0210* + FAS 1220	2
English	ENG 1301 + ENG 2301, ENG 2302, ENG 2303	6
Arabic	ARA 1201, ARA 1202, ARA 1203, ARA 3299, ARB 1201, ARB 1202, ARB 1203, ARB 1241	2
French	FRN 3210	2
Communication	COM 1301	3
History or Political Science	HIS 1301, HIS 2301, PSC 2301, HUM 2302, HUM 1310	3

(Walid Gouza, Dahbi Skali Mohammed)
Comment:
 When you refer to General Education Folder, create a subfolder for each discipline:
 Humanities, Social Science, ...

Humanities	LIT 2301, HUM 2305, HUM 2306, HUM 2307, PHI 2301, PHI 2302	3
Art Appreciation & Creation	HUM 2301, ART 1301, ART 1302, ART 1303, ART 1304, ART 1305, ART 3399, ENG 2320, COM 2327, LIT 3370	3
Social Sciences	ECO 1300, GEO 1301, PSY 1301, SOC 1301, SSC 1310	3
Civic Engagement	XXX ****SL, SLP 1101 or CIP 2000	1

(Maxime Cellier, Aboussikine Yasser) Comment: Make sure that each course picture is put in the right folder in the shared link	Area 2: Mathematics, Sciences and Engineering (30 SCH) Mathematics, Sciences and Engineering Requirements (26 SCH) <table border="1"> <thead> <tr> <th>Course Code</th><th>Course Name</th></tr> </thead> <tbody> <tr> <td>EGR 2302</td><td>Engineering Economics</td></tr> <tr> <td>MTH 1303</td><td>Calculus I: Differential and Integral Calculus</td></tr> <tr> <td>MTH 1304</td><td>Discrete Mathematics</td></tr> <tr> <td>MTH 2301</td><td>Calculus II: Multivariable Calculus</td></tr> <tr> <td>MTH 2320</td><td>Linear and Matrix Algebra</td></tr> <tr> <td>MTH 3301</td><td>Probability and Statistics for Engineers</td></tr> <tr> <td>PHY 1401</td><td>Physics I</td></tr> <tr> <td>PHY 1402</td><td>Physics II</td></tr> </tbody> </table> Basic Sciences Electives (4 SCH) <table border="1"> <thead> <tr> <th>Course Code</th><th>Course Name</th></tr> </thead> <tbody> <tr> <td>BIO 1401</td><td>Principles of Biology</td></tr> <tr> <td>BIO 1402</td><td>Environmental Biology</td></tr> <tr> <td>CHE 1401</td><td>Chemistry I</td></tr> </tbody> </table>	Course Code	Course Name	EGR 2302	Engineering Economics	MTH 1303	Calculus I: Differential and Integral Calculus	MTH 1304	Discrete Mathematics	MTH 2301	Calculus II: Multivariable Calculus	MTH 2320	Linear and Matrix Algebra	MTH 3301	Probability and Statistics for Engineers	PHY 1401	Physics I	PHY 1402	Physics II	Course Code	Course Name	BIO 1401	Principles of Biology	BIO 1402	Environmental Biology	CHE 1401	Chemistry I
Course Code	Course Name																										
EGR 2302	Engineering Economics																										
MTH 1303	Calculus I: Differential and Integral Calculus																										
MTH 1304	Discrete Mathematics																										
MTH 2301	Calculus II: Multivariable Calculus																										
MTH 2320	Linear and Matrix Algebra																										
MTH 3301	Probability and Statistics for Engineers																										
PHY 1401	Physics I																										
PHY 1402	Physics II																										
Course Code	Course Name																										
BIO 1401	Principles of Biology																										
BIO 1402	Environmental Biology																										
CHE 1401	Chemistry I																										
(Chahd Nezhari, Reda Heddad)	Chahd: check if (Oussama Oukkassou, Ilyas Bartal) work is correct. Report to me any anomalies Reda: check if (Walid Gouza, Dahbi Skali Mohammed) work is correct. Report to me any anomalies																										
(Mohammed Ismaili, Haytam Yazid) Comment: Make sure that each course picture is put in the right folder in the shared link	All Specializations (refer to the catalog to find each specialization courses)																										
(Yasmine Sour, Lina Bouyahyaoui)	Yasmine: check if (Maxime Cellier, Aboussikine Yasser) work is correct. Report to me any anomalies Lina: check if (Mohammed Ismaili, Haytam Yazid) work is correct. Report to me any anomalies																										

(Lina Oudghiri Idrissi Hassani, Ahmed Reda Benyaich) Comment: Make sure that each course picture is put in the right folder in the shared link	<table border="1"> <thead> <tr> <th>Course Code</th><th>Course Name</th></tr> </thead> <tbody> <tr> <td>CSC 1401</td><td>Computer Programming</td></tr> <tr> <td>CSC 2302</td><td>Data Structures</td></tr> <tr> <td>CSC 2305</td><td>Computer Organization and Architecture</td></tr> <tr> <td>CSC 2306</td><td>Object Oriented Programming</td></tr> <tr> <td>CSC 3315</td><td>Languages and Compilers</td></tr> <tr> <td>CSC 3323</td><td>Analysis of Algorithms</td></tr> <tr> <td>CSC 3324</td><td>Software Engineering</td></tr> <tr> <td>CSC 3326</td><td>Database Systems</td></tr> <tr> <td>CSC 3351</td><td>Operating Systems</td></tr> <tr> <td>CSC 3371</td><td>Computer Communications and Networks</td></tr> <tr> <td>CSC 3374</td><td>Advanced and Distributed Programming Paradigms</td></tr> <tr> <td>EGR 4300</td><td>Internship</td></tr> <tr> <td>EGR 4402</td><td>Capstone Design</td></tr> </tbody> </table>	Course Code	Course Name	CSC 1401	Computer Programming	CSC 2302	Data Structures	CSC 2305	Computer Organization and Architecture	CSC 2306	Object Oriented Programming	CSC 3315	Languages and Compilers	CSC 3323	Analysis of Algorithms	CSC 3324	Software Engineering	CSC 3326	Database Systems	CSC 3351	Operating Systems	CSC 3371	Computer Communications and Networks	CSC 3374	Advanced and Distributed Programming Paradigms	EGR 4300	Internship	EGR 4402	Capstone Design
Course Code	Course Name																												
CSC 1401	Computer Programming																												
CSC 2302	Data Structures																												
CSC 2305	Computer Organization and Architecture																												
CSC 2306	Object Oriented Programming																												
CSC 3315	Languages and Compilers																												
CSC 3323	Analysis of Algorithms																												
CSC 3324	Software Engineering																												
CSC 3326	Database Systems																												
CSC 3351	Operating Systems																												
CSC 3371	Computer Communications and Networks																												
CSC 3374	Advanced and Distributed Programming Paradigms																												
EGR 4300	Internship																												
EGR 4402	Capstone Design																												
(Yassmine Najib, Houda Bachr) Comment: Make sure that each course picture is put in the right folder in the shared link	Refer to the catalog to get the courses required for the Business Administration Minor.																												
(Tiphany Wairimu Muchemi, Patrick Nathan Nguea Jr) Comment: Make sure that each course picture is put in the right folder in the shared link	Refer to the catalog to get the courses required for the Communication Minor.																												
(Fatima Zahra Dakir, Akhaddar Adam) Comment:	Refer to the catalog to get the courses required for the Psychology Minor.																												

Make sure that each course picture is put in the right folder in the shared link	
(Nizar El Hamlaoui, Yassine Mallass) Comment: Make sure that each course picture is put in the right folder in the shared link	Refer to the catalog to get the courses required for the International Studies Minor.
(Zineb Berrada, Mariam Bouja)	Check if (Lina Oudghiri Idrissi Hassani, Ahmed Reda Benyaich) work is correct. Zineb, report to me any anomalies
(Fahd Rhoulam, El Ghali Mammam El Bourachdi)	Check if (Yassmine Najib, Houda Bachr) and (Tiphany Wairimu Muchemi, Patrick Nathan Nguea Jr) works are correct. Ghali, report to me any anomalies
(Nasser Mhasser, Ayat Elabidi)	Check if (Fatima Zahra Dakir, Akhaddar Adam) and (Nizar El Hamlaoui, Yassine Mallass) works are correct. Nasser, report to me any anomalies
(Achniouine Ikram)	You fill in two folders: Computing Electives (found in Computer science core and computing elective) and Electives. For the elective folder use the following (Add 10 random courses that can be counted as elective): A BSCSC major must complete 6 SCH of free electives. Free Elective 1* Free Elective 2* * Free electives can be selected in collaboration with and approval from the academic advisor and academic coordinator to reflect the student's personal and career interests. Graduate courses in SSE may be taken as electives with special permission from the Dean. The following courses are not allowed as free electives: CHE 1400, CSC 1300, GBU 3311, MTH 1305, MTH 1388, and PHY 1400. ☛ I will check the content of the two folders

Part 2: ChatGPT and MongoDB Time:

- ☒ Show that MongoDB is up and running; name your database properly
- ☒ Explain the choice of collections you will adapt in MongoDB.
- ☒ Explain the code used to create the collections
- You need to use ChatGPT to generate a JSON File for each picture.

- Each JSON must have **at least** the following properties: Course Code, Course Title, Number of Credits, Course Description, Course Discipline and pre OR co requisite(s)
 - Use Python and write a script that will be used to generate a JSON file for each course
 - ☒ **Explain the script used**
 - You need a validation agent that checks if the picture and the corresponding JSON file are correct!
 - ☒ **Show your work**
 - ☒ **List how many documents you have in your database**

Part 3: Neo4j and Cypher Time!

- ☒ **Show that Neo4j is up and running; name your database properly**
- Write a Python program that will be used to migrate MongoDB data to Neo4j database
 - ☒ **Show your work**
 - ☒ **Show the constructed graph and explore some random nodes**
- **Go to Neo4j browser**, and use the right cypher queries that will answer the following (☒ **Show your work and results**):
 - List all CSC3331 properties!
 - How many math courses do I need to take in the CS major? What are they?
 - List a scenario of courses I can take if I choose AI specialization? Make sure that you list the pre-requisites.
 - How many credits will I earn by the time I finish the bachelor's degree?
 - List the courses to take for a minor X
 - Remember the minor is 5 courses!
 - You can give two scenarios of courses to take!
 - What are the courses that I can not take as electives?
 - Suggest 3 courses that I can take as electives.
 - Are there any art courses related to drawing? List them along with their pre-requisite(s)
 - List a Computing Elective that that I can take as part of my major.
 - List all courses that have MTH1303 as a pre-requisite (keep an eye on these courses). Remove MTH1303 and replace it with two courses: MTH1311 and MTH1312
 - MTH1311 is a prerequisite for MTH1312
 - Courses that used to have MTH1303 as a pre-requisite will end up having MTH1311 as pre-requisite

- EXCEPTION MTH2301 will have MTH1312 as a pre-requisite
- Move EGR2302 (Engineering Economics) to Social Sciences courses
- List the pre-requisites for FIN3301
- List all the social science courses that I can choose from
- List How many credits will I earn by the time I finish the bachelor's degree? => should be the same as up!

You are done!